

NAMAN TYAGI

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SUMMARY

AI Engineer who owns systems end-to-end, from architecture through deployment to the stakeholders who use them daily. Hands-on experience building RAG pipelines, agentic workflows (LangGraph), fine-tuned models, and a self-hosted LLM gateway with failover, guardrails, and cost controls, then bridging them into real business environments. NYU MS CS graduate, May 2026. Published at IEEE (2024).

TECHNICAL SKILLS

Programming:	Python, Java, C/C++ , SQL, JavaScript
Agentic AI Systems:	LangGraph (Agentic Workflows), LangChain, Prompt Engineering, Tool-Use Design
ML Training & Evaluation:	Transformer Fine-Tuning (LoRA/PEFT), TensorFlow, PyTorch, Model Evaluation, RAG, Vector Databases (FAISS, ChromaDB, pgvector), Hugging Face
LLM Infrastructure:	FastAPI, PostgreSQL, Redis, Caching, Circuit Breakers, Load Testing (Locust), Prometheus, Grafana, Jaeger
Distributed Systems:	Apache Kafka, Apache Spark, Microservices, Event-Driven Architecture, REST APIs
Cloud & Infra:	AWS (Lambda, S3, DynamoDB, API Gateway), Azure, GCP, Docker

PROFESSIONAL EXPERIENCE

Software Engineer — Faculty Operations Platform Jan 2026 – Present
NYU Office of Faculty Affairs, NY

- Owning end-to-end delivery of an AI-powered FAQ chatbot grounded in retrieval, iterating on response quality and reliability based on real usage, alongside a records management system and workflow automation engine serving 500+ faculty.
- Designing automated, event-driven workflow pipelines to digitize manual, cross-departmental processes (compensation, hiring, approvals), ensuring correct routing and compliance checks across the full approval chain.
- Architecting backend services for structured record classification, converting legacy unstructured documents into searchable, queryable datasets via API-driven pipelines.

Software Engineering Intern (AI & BI Tools) Jun 2025 – Aug 2025
Mast-Jägermeister US, Inc., White Plains, NY

- Built production-grade internal tools combining backend ingestion pipelines, semantic search, and interactive dashboards for 50+ field team members across 1,000+ enterprise documents.
- Designed automated document processing pipelines with semantic indexing, reducing information retrieval time by ~70% compared to manual workflows.

AI Research Intern (Remote) Sep 2023 – Aug 2024
University of Essex, UK

- Conducted NLP research for early BDD detection, achieving a 12% F1-score improvement through transformer fine-tuning on 10K annotated clinical text samples; automated evaluation workflows, reducing experiment iteration time by ~40%.

Global Technical Support Engineer Jan 2024 – Jun 2024
Ciena, Gurugram, India

- Performed data-driven diagnostics on optical networks for 15+ global telecom clients (99.5% uptime SLAs); authored automation scripts reducing case resolution time by ~25%.

PROJECTS

Prism: LLM Gateway & Control Plane FastAPI, PostgreSQL, Redis, Docker, Prometheus, Grafana

- Built a self-hosted proxy layer between applications and LLM providers (OpenAI, Anthropic, local Ollama) with automatic failover, PII/prompt-injection guardrails, semantic + exact caching, and per-team cost tracking; OpenAI SDK-compatible via a one-line base_url swap.

- Engineered a hand-rolled circuit breaker (closed/open/half-open) and fallback chain; load tested to ~245–259 req/s (p50 ~25ms, p95 ~80ms) with 0% error rate during simulated provider outages, and fixed a Redis pool-exhaustion bug that had caused a ~77% error rate.

RAGBase: Enterprise Document Q&A System Python, LangChain, FAISS, Streamlit

- Architected a modular offline RAG system with configurable embedding models, chunking strategies, and vector search, returning cited, retrieval-grounded answers across 1,000+ documents with sub-2-second latency.

Sign Language Recognition Using CNN Python, TensorFlow, OpenCV, CNNs — Published IEEE 2024

- Built a custom CNN achieving 94%+ accuracy for gesture-to-text translation with skeleton-plotted augmentation; published in IEEE Xplore demonstrating cross-background generalization.

EDUCATION

NYU Tandon School of Engineering — M.S. Computer Science Aug 2024 – May 2026

ML, Computer Vision, Big Data, Cloud Computing, AI Resilience & Governance, Blockchain Security, Application Security, Algorithms

Amity University Noida — B.Tech Computer Science (Honors in AI & ML) Jul 2020 – Jun 2024

Exchange: Birkbeck University, London (Spring 2023) | Adelphi University, New York (Summer 2023)

Certifications: Google Cybersecurity Certificate | Google Cloud Big Data & ML | AWS, GCP & Azure Cloud Credentials